INCH-POUND

MIL-DTL-5184C
18 July 2016
SUPERSEDING
MIL-I-5184B
10 July 1989

DETAIL SPECIFICATION

INDICATOR, POSITION

Reactivated after 18 July 2016 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Comments, suggestions, or questions on this document should be addressed to Air Force Sustainment Center - Oklahoma City/AFLCMC/LZPES, 3001 Staff Drive, Suite 1AB81A, Tinker AFB, OK 73145 or emailed to ocalc.dsp@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6610

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1. SCOPE

1.1 Scope. This specification covers the requirements for one type of 28V dc, wing flap position indicator.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of these lists, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-L-3891	-	Luminescent Material and Equipment
MIL-T-5583	-	Transmitter, Position, 28 Volt DC
MIL-I-7062	-	Indicators, Position, Control Surfaces, 28 Volt DC, General Specification for
MS28023	-	Indicator-Position, Wing Flaps, Percent Marking
MS33639	-	Case, Instrument, Clamp-Mounted, Aircraft

(Copies of these documents are available at http://quicksearch.dla.mil.)

AIR FORCE/NAVY (AN)

AN5786	-	Transmitter-Position, 28 Volt DC
AN5787	-	Transmitter-Position, Gearless

(Copies of these documents are available at http://quicksearch.dla.mil.)

2.3 Order of Precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Indicators furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.1 and 6.3).

3.2 General. The requirements specified in the Specifications MIL-I-7062 are applicable as requirements of this specification. Additional requirements shall be as specified herein.

a. In the event the requirements of the general specifications and this specification conflict, the requirements of this specification shall govern.

3.2.1 Design and construction. The indicator shall be designed for operation with a position transmitter in accordance with Specification MIL-I-5583. The indicator pointer shall travel approximately 180 mechanical degrees for 90 degrees rotation of a transmitter conforming to Drawing AN5786 or 180 degrees rotation of a transmitter conforming to Drawing AN5787.

3.2.2 Resistance. The direct-current resistance of the indicator measured between the connector pins A, B, and C shall be 475 ±100 ohms. Deviation from the average of resistance values between the three coils shall not exceed ±1.50 percent.

3.2.3 Pointer overlap. The pointer length shall be such that the pointer tip will extend into the scale a distance equal to 0.333 to 0.67 of the length of the shortest graduations.

3.2.4 Electrical connector. A connector as shown on Specification MS28023 shall be furnished, unless otherwise approved for hermetically sealed instruments by the contracting services.

3.2.5 Cover glass position. The distance between the outer surface of the cover glass and the front edge of the case shall not exceed 0.031 inch.

3.2.6 Case. The case shall be in accordance with Specification MS28023. The case shall provide a hermetically sealed enclosure for the internal mechanism of the indicator.

a. The design of the indicator case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed. This shall be accomplished without the use of specific tools and fixtures, unless they are approved by the Contracting Services. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials which will be affected by the action of any atmosphere to which the indicator may be subjected.
b. The case shall be filled with helium of at least 98-percent purity, free of dust particles, and containing not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The pressure of the helium in the case shall be 1 atmosphere ±psi as referenced to an absolute vacuum.

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3.3 Dial. The dial shall conform to Figure 1. The dial shall be securely fastened to the frame of the mechanism in such a manner that it will not loosen or turn when the indicator is vibrated. The diameter of the dial shall be not shall be not less than 1.75 inches when measured across the outside ends of the graduations.

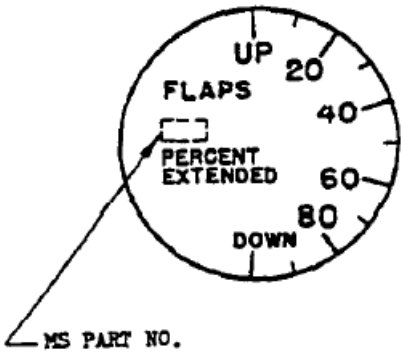
3.4 Markings. The dial markings shall be as shown on Figure 1, and shall be painted with a pale yellow, fluorescent luminescent material in accordance with MIL-L-3891. The painted surfaces shall experience no degradation as a result of the environmental tests in Section 4 of this specification.

3.5 Outline dimensions. The outline dimensions of the indicator shall conform to Specification MS28023.

3.6 Identification of product. The nameplate shall be marked with the following information:

INDICATOR, POSITION
MS Part No.
Stock No.
Manufacturer’s Part No.
Manufacturer’s Serial No.
Contract or Order No.
Manufacturer’s Name or Trade-Mark
U.S. Property

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MARKINGS	HEIGHT OR LENGTH INCHES ±0.015	WIDTH OF LINE OR GRADUATION INCH ±0.005	MATERIAL OR FINISH
Numerals 20, 40, 60, 80, and lettering “UP”	0.156	0.010	
Lettering “FLAPS”	0.125	0.010	
Lettering “DOWN”	0.109	0.010	Fluorescent
Lettering “PERCENT EXTENDED”	0.093	0.010	Luminescent
Major Graduations	0.187	0.031	
Minor Graduations	0.043	0.020	
Shaded Portion of Pointer			
MS Part No.	0.062	0.020	Durable
Background of Dial			Dull Black

FIGURE 1. Dial Markings.

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3.7 Recycled, recovered, or environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.1.1).
b. Conformance inspection (see 4.1.1 and 4.1.2).

4.1.1 General. Methods of sampling, inspection, and tests shall be as specified in Specification MIL-I-7062 and as follows:

4.1.2 Scale Error Test. The scale error shall be read from the test transmitter scale with the indicator pointer on each major graduation of the indicator dial.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The flaps position indicator covered by this specification is intended to indicate the position of the wing flaps in percent of travel from the up position. The position transmitters used with this indicator are connected to the flap actuation mechanism so that the transmitter shaft is rotated 90 or 180 mechanical degrees as applicable for 100-pecent flap travel. The indicator pointer rotates approximately 180 degrees to indicate the full down position.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Packaging requirements (see 5.1).

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-5184 whether or not such products have actually been listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Air Force Sustainment Center – Oklahoma City/AFLCMC/LZPES, 3001 Staff Drive, Suite 1AB81A, Tinker AFB, OK 73145 or email to ocalc.dsp@us.af.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Subject term (key word) listing.

Cover glass position
Pointer overlap
Transmitter

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - AV
Navy - AS
Air Force - 71

Preparing activity:
Air Force - 71

(Project 6610-2016-003)

Reviewer activities:
Army - MI
Air Force - 11
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